

Baseline-Relative Predictability Horizons for Cross-Domain Forecast Evaluation

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Abstract

Forecasting studies are typically summarized at a few fixed lead times, which can obscure the practical question of how long a method remains useful relative to an explicit baseline. We recast predictability-horizon estimation as a forecast evaluation task and measure baseline-relative performance through the forecast skill curve $\mathbf{Skill}(h)$, obtained using a leakage-free rolling-origin procedure with persistence as the reference. To distinguish sporadic from sustained benefits, we report two complementary horizon descriptors: $\mathbf{H}^*(\mathbf{relax})$, the maximum evaluated horizon such that $\mathbf{Skill}(h) > \mathbf{0}$ (allowing intermediate non-positive gaps), and $\mathbf{H}^*(\mathbf{strict})$, the length of the longest contiguous interval of positive skill. When needed (especially for fragmented or late-recovery profiles), we additionally report interval locations and sign-change structures. Applied to four public datasets, the descriptors reveal different skill profiles: a short positive window in electric load, early positive islands followed by a longer positive interval in PM_{2.5}, sustained positive skill in wind speed, and fragmented positive intervals in traffic speed. A fixed seasonal-persistence sensitivity, evaluated on exact common support, changes strict continuity and interval location in several domains even when relaxed reach is unchanged, showing that the descriptors depend on the baseline. These patterns motivate reporting multiple operational descriptors beyond fixed-horizon error summaries so that total positive reach, sustained contiguous usefulness, and fragmented interval structure can be distinguished under a shared leakage-free protocol.

Keywords: forecast evaluation; predictability horizon; baseline-relative skill; rolling-origin evaluation; cross-domain forecasting

1 Introduction

In applied forecasting, evaluation is often reported at a small number of fixed lead times. This practice can obscure when (and for how long) a method is operationally useful relative to a strong baseline, especially when multi-step performance is non-monotonic or fragmented across the horizon. Methodological audits also document recurrent evaluation failures—including temporally invalid partitioning, leakage-prone preprocessing, omitted parsimonious baselines, and uncritical metric selection—that can materially distort conclusions about practical forecasting value (Hewamalage et al. 2023).

A remaining evaluation problem is how to quantify baseline-relative usefulness over the full forecast horizon under a shared, leakage-free protocol. We address this problem through rolling-origin evaluation, train-only preprocessing, and persistence as an explicit reference baseline. We report performance as a horizon-wise skill curve, $\text{Skill}(h)$, and use two descriptors to distinguish the last horizon with positive skill from the longest uninterrupted positive interval. We apply this design to four domains—PM_{2.5} air quality, electric load, wind speed, and traffic speed—chosen to span heterogeneous dynamics and baseline competitiveness.

We frame predictability-horizon estimation as an evaluation-methodology contribution for forecasting systems in machine learning and data mining. The goal is not to create another applied leaderboard, but to establish a protocol for characterizing and comparing useful predictive reach by horizon relative to a clearly specified baseline, thereby complementing recent methodological work on multi-step forecasting (Green et al. 2025; Kreuzer et al. 2025).

The paper makes three contributions:

1. **A reproducible leakage-free evaluation framework:** we specify a rolling-origin protocol with train-only preprocessing that computes horizon-wise skill $\text{Skill}(h)$ relative to an explicit persistence baseline, enabling comparable operational assessments across domains.
2. **Operational horizon descriptors for useful predictive reach and continuity:** we instantiate the generic predictability-horizon concept via two complementary descriptors— $H^*(\text{relax})$, the maximum evaluated horizon such that $\text{Skill}(h) > 0$ (allowing intermediate non-positive gaps), and $H^*(\text{strict})$, the length of the longest contiguous positive-skill interval $[h_{\text{start}}, h_{\text{end}}]$ —augmented, when needed, with interval locations and sign-change structure for fragmented or late-recovery profiles.
3. **A cross-domain empirical analysis under a shared protocol:** we show that domains differ not only in positive reach but also in continuity and fragmentation, and we quantify how the descriptors change under a fixed alternative baseline on exact common support.

Together, these contributions provide a leakage-free, baseline-relative basis for estimating and comparing operational predictability horizons across domains.

2 Related Work

2.1 Forecast evaluation metrics

Within the forecasting literature, forecast evaluation is most often framed in terms of accuracy metrics such as MAE and RMSE, along with scaled or relative variants that enable comparisons across different series and applications. [Hyndman and Koehler \(2006\)](#) survey the most commonly used measures and underscore that no single metric is universally ideal; the appropriate choice depends on scale, loss asymmetry, and the decisions the forecasts are intended to inform ([Petropoulos et al. 2022](#)).

For probabilistic forecasts, evaluation is commonly grounded in strictly proper scoring rules, which ensure that the expected score is optimized by the forecaster’s true predictive distribution ([Gneiting and Raftery 2007](#)). This framework shifts the evaluation focus from point accuracy to the full predictive distribution, but the predominant empirical convention remains point-forecast-based reporting at fixed horizons.

When comparing forecasting methods, statistical tests of predictive accuracy (e.g., the Diebold–Mariano framework) are also frequently used to assess whether observed differences are larger than would be expected by chance ([Diebold and Mariano 1995](#)). This work does not employ formal significance tests; instead, skill is assessed relative to explicit baselines across the full set of evaluated horizons, with baseline sensitivity evaluated on exact common support.

Empirical work typically reports errors for one or a few fixed horizons. While convenient, this fixed-horizon focus can hide how performance changes with lead time, including non-monotonic, fragmented, or delayed skill profiles in multi-step settings. Operationally, it can therefore be unclear whether a method is unhelpful overall or whether it provides baseline-relative usefulness only over specific horizon intervals. Fixed-horizon reporting also makes it difficult to directly identify where horizon-wise skill becomes non-positive relative to an explicit baseline.

Large-scale benchmark exercises have reinforced this convention. Comparisons such as the M4 competition ([Makridakis et al. 2020](#)) and repositories like the Monash Time Series Forecasting Archive ([Godaheva et al. 2021](#)) assess many methods over many series, but usually at fixed or pre-chosen horizons. This setup is effective for contrasting methods on relative accuracy; yet, it does not directly characterize how forecast skill evolves over the full horizon range or at which point baseline-relative skill is exhausted.

A further drawback of fixed-horizon summaries is that, even when a strong baseline is available, they only indirectly address operational relevance. From an applied standpoint, the key question is often not whether a method lowers error at a single, predefined horizon, but whether it consistently improves on a baseline across the set of horizons that matter for planning, control, or service-level agreements. [Hyndman and Athanasopoulos \(2018\)](#) consolidates prevailing guidance on forecast evaluation and highlights the need for out-of-sample validation and for matching the evaluation design to how forecasts are generated and used. This points to evaluation constructs that summarize performance over horizons and that are interpretable relative to a clearly specified reference forecast. Related methodological audits emphasize that common evaluation pitfalls (e.g., leakage-prone splits and missing strong baselines)

can materially distort conclusions about practical forecasting value (Hewamalage et al. 2023); we use this as motivation for the leakage-free, baseline-relative protocol described in Section 4.

2.2 Forecast skill evaluation

Baseline-relative evaluation is often expressed in terms of forecast skill, which measures the error of a candidate forecasting method against that of a reference forecast:

$$\text{Skill}(h) = 1 - \frac{E_{\text{model}}(h)}{E_{\text{baseline}}(h)}. \quad (1)$$

This representation is fundamental in forecast verification, where the quality of a forecast is defined only in relation to a baseline (Murphy 1993). In practical forecasting settings, the same rationale underpins operational interpretation: a model that marginally improves absolute error may still be operationally unimportant if it does not surpass a strong baseline at the horizons that matter, whereas a model whose gains appear only after a period of short-horizon baseline dominance can still be highly useful for longer-range decision-making.

Skill-based reporting aligns naturally with multi-horizon evaluation because it produces a horizon-specific quantity, the skill curve $\text{Skill}(h)$, instead of a single aggregate metric. This facilitates distinguishing between evaluation outcomes driven by the strength of the baseline and those stemming from modeling decisions. It also underscores the importance of rigorous temporal evaluation protocols. Out-of-sample forecast tests are typically implemented via time-aware designs such as rolling-origin evaluation, which repeatedly re-fits models and scores forecasts as the forecast origin moves forward in time (Tashman 2000). These designs mirror real forecasting practice and mitigate the risk that conclusions hinge on a particular train–test split. The suitability of rolling-origin designs, in contrast to blocked or random cross-validation for time series, has been investigated empirically, with evidence indicating that respecting temporal order is crucial for reliable performance estimation (Cerqueira et al. 2020; Bergmeir and Benítez 2012).

In machine-learning-based forecasting, this principle is coupled with a further practical issue: preventing temporal leakage arising from preprocessing, feature engineering, hyperparameter tuning, or validation procedures that inadvertently exploit future information (Kapoor and Narayanan 2023). Such leakage can artificially boost apparent skill, especially at short horizons where persistence effects are strong and where small misalignments or preprocessing-related leakage can dominate evaluation results. Therefore, leakage-free temporal validation is essential for interpreting skill curves and for using baseline-relative evaluation to draw operational conclusions about when forecasts are genuinely beneficial.

2.3 Predictability limits

Finite predictability in complex systems has long been recognized as a core issue in forecasting and dynamical systems. Lorenz (1969) demonstrated that sensitivity to initial conditions can create inherent constraints on the lead times over which accurate

prediction is achievable. In empirical forecasting, similar limitations appear as the horizon grows and the baseline becomes increasingly difficult to surpass. However, these limits are not solely characteristics of a particular model class; they emerge from the interplay among the system’s dynamics, the choice of reference forecast, and the evaluation protocol.

Fixed-horizon accuracy metrics remain the dominant reporting convention (Makridakis et al. 2020; Godahewa et al. 2021), but they do not directly identify the horizon at which baseline-relative skill is exhausted. From an operational standpoint, the key quantity is the range of horizons over which a method achieves positive skill relative to a baseline, under a clearly specified, temporally coherent evaluation design.

Cross-domain comparability is an evaluation challenge because reported horizon-wise skill depends on baseline choice, preprocessing, evaluated support, and temporal validation design. Baseline-relative skill measures and time-aware evaluation are well established (Hyndman and Koehler 2006; Murphy 1993; Tashman 2000). Large benchmark studies (Makridakis et al. 2020; Godahewa et al. 2021) provide broad accuracy comparisons, whereas the present analysis addresses a different operational question: how baseline-relative skill varies across the evaluated horizon under a specified common protocol.

Without a common evaluation protocol, differences in reported predictability across domains cannot be cleanly separated from differences in baseline specification, preprocessing, evaluated support, or validation design. This evaluation problem motivates baseline-relative horizon descriptors that summarize the span of positive skill in a way that is robust to fragmented or slowly emerging skill profiles and that supports comparison across forecasting systems without conflating predictability with methodological novelty or dataset-specific tuning.

3 Problem Formulation

Consider a time series y_t , $t = 1, \dots, T$. A forecasting model generates predictions

$$\hat{y}_{t+h} = f_{\theta}(y_t, \dots, y_{t-L+1}), \quad (2)$$

where L denotes the input window length and h the forecast horizon. The corresponding forecast error is

$$e_{t,h} = y_{t+h} - \hat{y}_{t+h}. \quad (3)$$

At each horizon h , forecast errors across evaluation origins are aggregated into an error measure $E(h)$, such as MAE or RMSE. Applying the same error measure to a baseline forecasting method $b(\cdot)$ yields the baseline error curve $E_{\text{baseline}}(h)$.

3.1 Operational Predictability Horizon

In this work, operational predictability is understood as the range of time over which a forecasting system exhibits positive skill relative to an explicit baseline, under a fixed, leakage-free evaluation setup.

We use H^* as the generic conceptual notation for the operational predictability horizon: the maximum forecast horizon at which a model exhibits positive skill relative to an explicit baseline under a specified evaluation protocol.

Formally, the generic *Forecast Skill Horizon* is

$$H^* = \max \{h \in \mathcal{H} : \text{Skill}(h) > 0\}, \quad \text{Skill}(h) = 1 - \frac{E_{\text{model}}(h)}{E_{\text{baseline}}(h)}. \quad (4)$$

Here, $\mathcal{H}$ denotes the set of evaluated forecast horizons. H^* is protocol-defined: its value depends on the baseline choice (persistence in the primary analysis), preprocessing strategy, error metric, evaluated support, and evaluation scheme (leakage-free rolling-origin). This dependency is intentional, as it makes the descriptor reproducible and operationally interpretable.

Operational definitions:

In empirical multi-step settings, $\text{Skill}(h)$ curves may be non-monotonic, fragmented, or exhibit delayed recovery, so the single generic definition above is best treated as a conceptual anchor. For example, a curve with early negative skill but a late positive recovery yields a large $H^*(\text{relax})$ but a short $H^*(\text{strict})$ —two descriptors that a single maximum-horizon summary conflates.

We therefore instantiate H^* through two complementary reported descriptors, defined over the evaluated horizon set $\mathcal{H} = \{1, 2, \dots, H_{\text{max}}\}$ and the positive-skill subset $\mathcal{H}^+ = \{h \in \mathcal{H} \mid \text{Skill}(h) > 0\}$. In the remainder of the paper, H^* is used as the generic conceptual label for the operational predictability horizon, whereas all empirical reporting is based on the two instantiated descriptors $H^*(\text{relax})$ and $H^*(\text{strict})$.

$H^*(\text{relax})$: The relaxed horizon is the maximum evaluated horizon such that $\text{Skill}(h) > 0$, allowing intermediate horizons with non-positive skill.

$$\begin{aligned} H^*(\text{relax}) &= \max(\mathcal{H}^+ \cup \{0\}), \\ \mathcal{R} &= \{h \in \mathcal{H} : h \leq H^*(\text{relax})\}, \\ \mathcal{G} &= \mathcal{R} \setminus \mathcal{H}^+ = \{h \in \mathcal{R} : \text{Skill}(h) \leq 0\}. \end{aligned} \quad (5)$$

The set $\mathcal{G}$ contains the non-positive gaps inside the relaxed reach and may be non-empty. These horizons are included in the span, not counted as horizons with positive skill. If $\mathcal{H}^+$ is empty, the relaxed horizon is zero and both $\mathcal{R}$ and $\mathcal{G}$ are empty.

$H^*(\text{strict})$: To capture sustained usefulness, let $\mathcal{I}$ be the collection of all contiguous integer intervals $I = [h_{\text{start}}, h_{\text{end}}]$ with $I \subseteq \mathcal{H}^+$. The strict horizon is the *length* of the longest such interval:

$$H^*(\text{strict}) = \max_{I \in \mathcal{I}} |I|, \quad |I| = h_{\text{end}} - h_{\text{start}} + 1. \quad (6)$$

If no positive-skill interval exists, we define $H^*(\text{strict}) = 0$. We additionally report the maximizing interval location $[h_{\text{start}}, h_{\text{end}}]$ and, for fragmented or late-recovery profiles, the full sign-change structure of $\text{Skill}(h)$. If several intervals tie for maximum

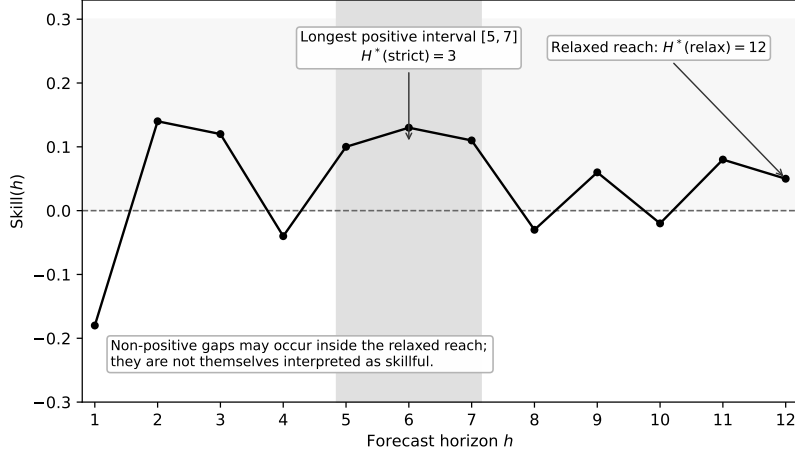


Fig. 1 Conceptual illustration of a baseline-relative skill curve $\text{Skill}(h)$ with both positive and negative regions. The relaxed horizon $H^*(\text{relax})$ is the maximum evaluated horizon with $\text{Skill}(h) > 0$, whereas the strict horizon $H^*(\text{strict})$ is the length of the longest contiguous positive-skill interval $[h_{\text{start}}, h_{\text{end}}]$.

length, the earliest interval is reported. $H^*(\text{time})$ denotes the wall-clock duration corresponding to $H^*(\text{strict})$ (not to $H^*(\text{relax})$), and it is reported alongside it.

Together, these two descriptors distinguish total positive reach from sustained contiguous usefulness — a distinction that matters in the fragmented and delayed-onset domains studied below. Figure 1 illustrates this distinction: relaxed reach extends to the last positive-skill horizon even when intervening gaps occur, whereas strict reach measures the longest uninterrupted positive-skill interval.

3.2 Operational interpretation and actionability

$H^*(\text{relax})$ describes the outer extent at which positive baseline-relative skill is still observed; it does not license forecasts at every preceding horizon. $H^*(\text{strict})$, reported together with $[h_{\text{start}}, h_{\text{end}}]$, identifies the longest continuous decision-relevant window in which the model improves on the chosen baseline. Fixed-horizon summaries cannot reveal delayed onset, intermediate gaps, fragmented positive islands, interval location, or late recovery. The two descriptors therefore answer different operational questions: how far any positive skill reaches, and where the longest continuous positive-skill window lies. Neither descriptor alone establishes that deployment is worthwhile. A practitioner must also consider the size and uncertainty of the improvement, the relevant decision loss, and the cost of using the forecast. The descriptors identify intervals to examine; they are not a substitute for that assessment.

4 Evaluation Framework

This section specifies the evaluation components used to compute horizon-wise baseline-relative skill curves $\text{Skill}(h)$ and the associated horizon descriptors: the reference baseline, error metrics, and the leakage-free rolling-origin validation protocol.

4.1 Baseline models

In every domain, we evaluate models against a persistence baseline. At forecast origin t , the baseline prediction for horizon h is simply the most recent observation available at time t . Persistence is intentionally adopted as a strong operational reference because it is easy to implement, transparent, and often highly competitive for short lead times (Murphy 1993; Hyndman and Athanasopoulos 2018). Accordingly, the evaluation objective is not just to report absolute error levels but to determine the horizons at which a forecasting method yields positive skill relative to this baseline.

To assess baseline sensitivity, we additionally pre-specify seasonal persistence before examining its results:

$$\hat{y}_{t+h}^{\text{season}} = y_{t+h-k_h s}, \quad k_h = \left\lceil \frac{h}{s} \right\rceil, \quad (7)$$

where $s = 24$ observations in the three hourly domains and $s = 7$ in daily electric load. The ceiling term selects the latest observation at the same seasonal phase as the target that is available no later than origin t , including when $h > s$. The period is fixed from sampling frequency and calendar structure, not selected from evaluation performance. For this sensitivity, LightGBM, persistence, and seasonal persistence are regenerated together, and comparisons are permitted only when forecast origin, target timestamp, horizon, and y_{true} match exactly. We report the number of records retained on this common support. At horizons that are exact multiples of s , this seasonal rule reduces to y_t and therefore coincides with ordinary persistence. Consequently, an unchanged relaxed endpoint at such a horizon is partly structural and is not interpreted as baseline invariance; changes in continuity and interval location remain informative.

4.2 Evaluation metrics

Forecast errors are evaluated primarily using mean absolute error (MAE) (Hyndman and Koehler 2006), which serves as the basis for computing baseline-relative skill at each horizon. Root mean squared error (RMSE) is additionally reported as a complementary error measure, but the main operational analysis throughout this work is framed in terms of $\text{Skill}(h)$ computed from MAE. This design choice ensures that horizon-specific comparisons remain straightforward to interpret: at each lead time, we can directly assess whether a given forecasting approach improves upon persistence.

4.3 Temporal validation protocol

The primary analysis adopts rolling-origin evaluation with train-only preprocessing and persistence as the explicit reference baseline. The alternative-baseline sensitivity

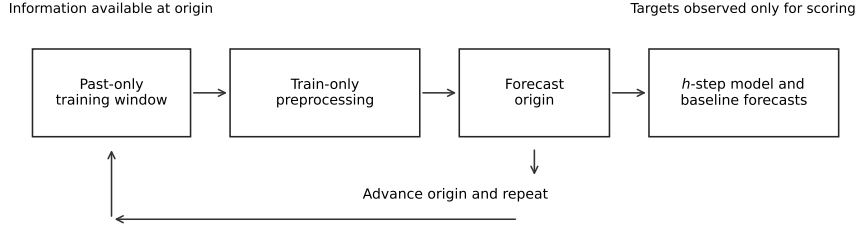


Fig. 2 Schematic of the leakage-free rolling-origin evaluation protocol. Each origin uses a past-only training window with train-only preprocessing, generates h -step-ahead forecasts for the model and the persistence baseline, and then rolls the origin forward in time.

follows the same temporal-validation principles and exact common-support alignment, with seasonal persistence as the additional reference. This design yields leakage-free, out-of-sample horizon-wise skill estimates that are interpretable relative to the specified baseline. Figure 2 shows how the training information set, forecast origin, and target horizons advance without exposing future observations to model fitting or preprocessing.

All experiments use strictly time-ordered, leakage-free evaluation. The procedure adopts a rolling-origin framework (Tashman 2000):

1. define a chronologically ordered training window,
2. fit any preprocessing steps using training data only,
3. produce model and persistence forecasts for horizon h ,
4. advance the forecasting origin,
5. iterate over all horizons and origins.

Random train-test splits are intentionally avoided because they may introduce temporal leakage. For all domains, model and baseline forecasts are generated and compared only at aligned timestamps.

5 Datasets

Table 1 provides an overview of the application domains, datasets, target variables, temporal resolution, and maximum forecast horizon.

The four domains are selected to support methodological comparison under a shared, leakage-free evaluation design rather than to provide exhaustive coverage of forecasting applications. They span heterogeneous temporal structures (from highly regular aggregate demand to intermittent traffic and wind dynamics), differ markedly in baseline competitiveness against persistence, and vary in temporal resolution and operational context (hourly environmental and mobility signals versus a daily energy series). Together, they form a diverse but controlled public cross-domain testbed for studying horizon-wise baseline-relative skill and its summaries.

Under this shared protocol, the four domains yield distinct baseline-relative skill-curve profiles, enabling the comparison of operational reach, contiguous usefulness, and (when present) fragmented interval structure.

Domain	Dataset	Variable	Frequency	$H_{\max}$
Air quality	UCI Beijing PM2.5 (2010–2014)	PM _{2.5}	hourly	48
Energy	UCI Electricity Load Diagrams 2011–2014	Aggregate load	daily	1
Wind	NREL Wind Toolkit	Wind speed	hourly	48
Traffic	PeMS / METR-LA	Traffic speed	hourly	72

Table 1 Public datasets used for cross-domain predictability analysis. In the energy domain, we use the UCI Electricity Load Diagrams 2011–2014 dataset (Trindade 2015). The aggregate daily series is obtained by summing all 370 client series and resampling to daily totals, starting from 2013-03-06, which is the first date with $\geq 90\%$ active clients over seven consecutive days.

In the traffic domain, the raw source variable corresponds to METR-LA sensor speed measurements (Li et al. 2018).

5.1 Data representation

For each domain, the cleaned data used in the experiments is standardized into a univariate format containing a timestamp column and a single target variable. In the electric-load domain, the cleaned representation is (`timestamp`, `value`) at daily resolution. The aggregate series is defined as the daily sum of all available client-level series from the stable-coverage start date onward. This explicit specification of the aggregation procedure promotes reproducibility and ensures alignment between data preparation and experimental evaluation.

6 Experimental Setup

6.1 Forecast horizons

The maximum forecast horizons are chosen to balance practical operational relevance with a sufficient length to capture the onset and progression of skill decay:

- PM_{2.5}: $H_{\max} = 48$,
- Electric load (daily): $H_{\max} = 1$,
- Wind: $H_{\max} = 48$,
- Traffic: $H_{\max} = 72$ (hourly).

When supported by operational considerations, these upper limits may differ across domains. The three hourly domains are evaluated for at least $h = 1, \dots, 48$, with traffic extended to $h = 72$, whereas the daily electric-load experiment is evaluated at $h = 1$.

6.2 Models

The empirical study compares forecasting performance against an explicit persistence baseline and employs LightGBM as the principal forecasting model. Evaluation semantics are shared across domains, while the fixed feature and estimator configurations follow the experiment scripts that produced the submitted results.

LightGBM is a decision tree-based gradient boosting framework that is particularly well suited for tabular, lag-based forecasting formulations (Ke et al. 2017). For the three hourly domains (PM_{2.5}, wind, and traffic), the fixed configuration is:

- input lags: $\{0, 1, 2, 3, 6, 12, 24, 48\}$ relative to the forecasting origin,
- rolling-origin stride: 24,
- number of estimators: 50,
- learning rate: 0.05; number of leaves: 31; subsample and feature fractions: 0.9; random seed: 42.

Lag 0 (y_t) is included so that the model and baseline share the same information at the forecasting origin; this does not introduce leakage since the target remains y_{t+h} for $h \geq 1$. The daily electric-load experiment uses a fixed configuration: 14 daily lags plus a weekly index-phase feature ($i \bmod 7$), an expanding past-only training set, 100 estimators, learning rate 0.05, 15 leaves, and random seed 42. Evaluation starts after 365 daily observations; this threshold is not the length of a rolling training window. The index-phase feature is not a timestamp-derived day-of-week label. These configurations were not selected by tuning on the evaluation period.

As a model-substitution check for wind and traffic only, we also evaluate an ARIMA(2, 0, 0) model, an autoregression of order two with a constant, under the same rolling-origin protocol, persistence baseline and training window as LightGBM. For each horizon h the model is fitted on the same horizon-shifted target and the h -step-ahead forecast is evaluated. The order (2, 0, 0) was fixed in advance as a low-order autoregression; it was not selected with ACF/PACF, information criteria or validation. We therefore treat this check as an illustration of descriptor sensitivity to model class, not as a tuned competitor, and report an *a posteriori* order diagnostic in Appendix A.2.

The persistence baseline is defined as the most recent observed value available at the prediction origin.

$$\hat{y}_{t+h}^{\text{base}} = y_t, \quad (8)$$

and is applied uniformly in all domains. This baseline is deliberately strong in operational forecasting contexts and serves as the reference level against which baseline-relative usefulness is assessed. Table 2 summarizes the baseline definition and an illustrative domain-specific example. Units follow the corresponding source metadata; for electric load, the aggregation script sums 15-minute kW readings, so division by four expresses the aggregate as kWh.

No hyperparameter tuning is carried out in any domain. The protocol is kept deliberately simple, without optimizing configurations on the evaluation period. The observed profiles nevertheless depend on the features and configuration as well as the dataset and baseline. The exact fixed configuration used in each domain is reported above and retained for reproducibility; no claim of identical feature construction across hourly and daily domains is made.

Domain	Target unit	Persistence rule	Illustrative explanation
PM _{2.5}	$\mu\text{g m}^{-3}$	$\hat{y}_{t+h}^{\text{base}} = y_t$	If $y_t = 42 \mu\text{g m}^{-3}$, every lead uses $42 \mu\text{g m}^{-3}$.
Load	Sum of 15-min kW readings	$\hat{y}_{t+h}^{\text{base}} = y_t$	The stored daily aggregate follows the producer script; dividing it by four gives the equivalent energy in kWh.
Wind	m s^{-1} at 100 m	$\hat{y}_{t+h}^{\text{base}} = y_t$	If $y_t = 6.8 \text{ m s}^{-1}$, every lead uses 6.8 m s^{-1} .
Traffic	mph	$\hat{y}_{t+h}^{\text{base}} = y_t$	If $y_t = 57$ mph, every lead uses 57 mph.

Table 2 Persistence definition and verified target units in each domain. Electric-load values are retained on the exact scale produced by the historical aggregation script; the stated conversion follows the source dataset’s 15-minute kW sampling.

6.3 Reproducibility rules

All tables and figures are generated directly from the experiments carried out under the stated evaluation protocol. To support replication, the data-processing and evaluation scripts operate on the cleaned, domain-specific datasets, with each step documented so that the reported results can be reproduced from the specified inputs.

7 PM_{2.5} Domain Results

7.1 PM_{2.5} experimental setting

The Beijing PM_{2.5} case (Chen 2015) uses hourly measurements, persistence, and the fixed hourly LightGBM configuration with $H_{\text{max}} = 48$. We assess performance through MAE-based skill. This example shows why the first positive horizon and the longest positive interval need not describe the same part of a skill curve.

7.2 Observed skill behavior

For PM_{2.5}, the Skill(h) curve in Figure 3 has three positive-skill intervals: [7, 11], [13, 14], and [27, 48], separated by non-positive regions. The reported LightGBM curve therefore yields $H^*(\text{relax}) = 48$, $H^*(\text{strict}) = 22$, and a longest positive interval $[h_{\text{start}}, h_{\text{end}}] = [27, 48]$, with $H^*(\text{time}) = 22$ h and five sign changes.

7.3 Interpretation

Positive skill at $h = 48$ sets the relaxed endpoint. The relaxed span therefore covers all 48 horizons, including the 19 with non-positive skill between and before the positive intervals; only 29 horizons have positive skill. The strict descriptor instead selects the 22-hour interval [27, 48], not either of the shorter early intervals. PM_{2.5} thus combines fragmentation with a longer late positive interval; it does not show positive skill only at long leads.

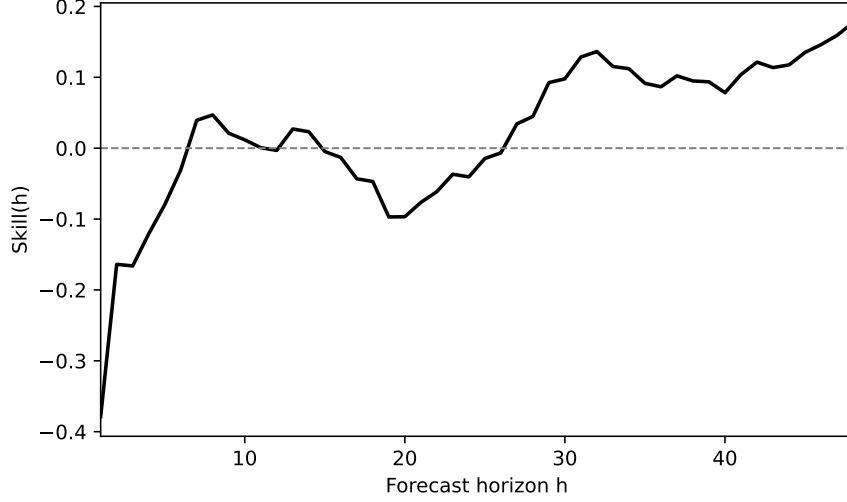


Fig. 3 PM_{2.5} LightGBM skill curve. Short positive intervals at $[7, 11]$ and $[13, 14]$ precede the longest positive interval, $[27, 48]$. The relaxed endpoint is 48 hours and the strict interval length is 22 hours.

8 Electric Load Domain Results

8.1 Energy experimental setting

The second domain considered is aggregate daily electricity demand, derived from the UCI Electricity Load Diagrams 2011–2014 dataset. This experiment uses daily observations, a persistence baseline, and the canonical comparable LightGBM protocol, with performance evaluated through MAE-based skill. The energy setting is especially informative because the series is highly regular, and persistence is expected to be particularly difficult to outperform; more broadly, electric load forecasting is a canonical operational setting where benchmarking and uncertainty-aware evaluation are central concerns (Hong and Fan 2016).

8.2 Observed skill behavior

In the canonical reported setting, daily electric load is operationally summarized at $h = 1$, yielding $H^*(\text{relax}) = 1$ and $H^*(\text{strict}) = 1$ with $[h_{\text{start}}, h_{\text{end}}] = [1, 1]$. For visualization purposes, Figure 4 additionally shows an auxiliary extended-horizon view up to $h = 7$ under the same protocol, highlighting that positive skill against persistence decays rapidly and remains confined to a very short early interval. On the primary one-day range, the two descriptors coincide:

$$H^*(\text{relax}) = 1, \quad H^*(\text{strict}) = 1, \quad [h_{\text{start}}, h_{\text{end}}] = [1, 1], \quad H^*(\text{time}) = 1 \text{ day}. \quad (9)$$

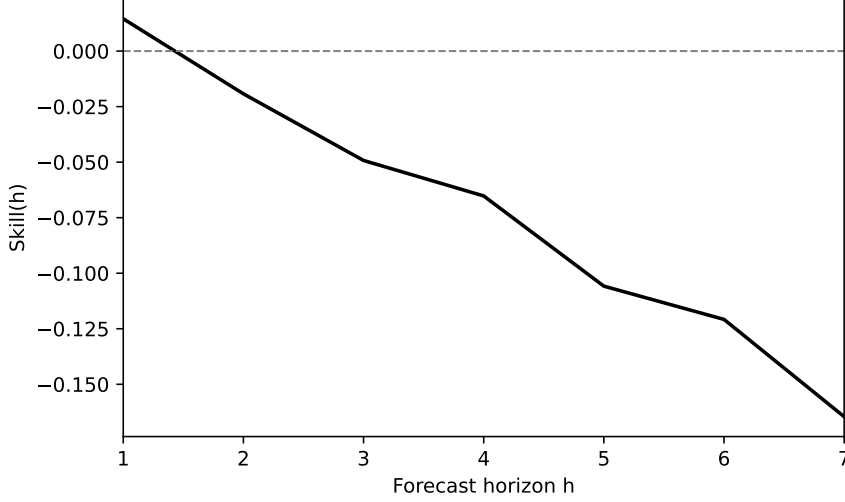


Fig. 4 Electric-load skill curve $\text{Skill}(h)$ under the leakage-free rolling-origin protocol. The canonical cross-domain analysis uses $H_{\max} = 1$, yielding $H^*(\text{relax}) = 1$ and $H^*(\text{strict}) = 1$ at $h = 1$. The extended view up to $h = 7$ is shown only to illustrate the rapid decay of positive skill against persistence beyond the first day.

8.3 Interpretation

With only $h = 1$ in the primary analysis, the two descriptors coincide by construction: the one evaluated day has positive skill. This result alone cannot establish what happens beyond one day. The auxiliary view in Figure 4 illustrates the longer-range behavior and must be distinguished from the one-day cross-domain summary.

9 Wind Domain Results

9.1 Wind experimental setting

The third domain considered is the NREL Wind Toolkit (Draxl et al. 2015), which provides hourly wind-speed time series. The wind experiment uses hourly observations, a persistence baseline, and the canonical comparable LightGBM protocol under $H_{\max} = 48$, with performance again evaluated through MAE-based skill. This domain is included to assess whether the proposed horizon descriptors also characterize settings in which positive skill is sustained across most or all of the evaluated horizon range.

9.2 Observed skill behavior

For wind, positive skill is obtained over the entire set of evaluated horizons, yielding a uniformly positive skill curve across $h = 1, \dots, 48$ (Figure 5). This shape indicates comparatively weak baseline competitiveness: persistence is outperformed not only at longer lead times but also at the shortest horizons. Under the operational variants

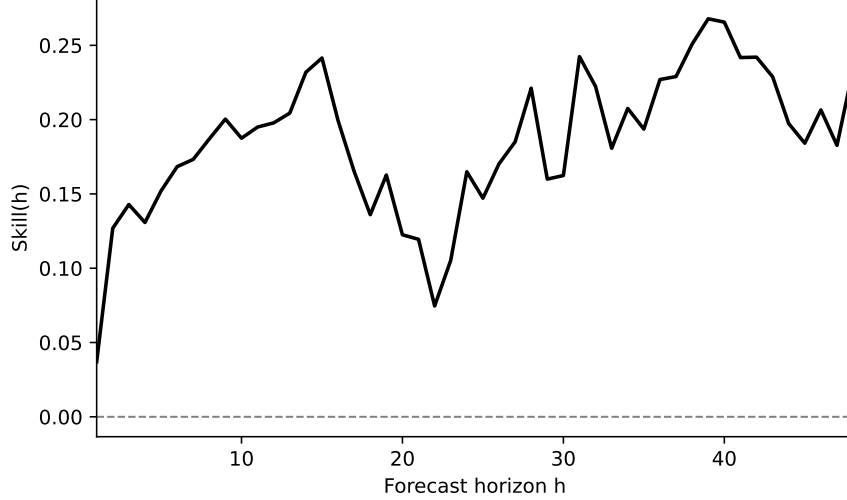


Fig. 5 Wind skill curve $\text{Skill}(h)$ under the leakage-free rolling-origin protocol. The skill-profile pattern shows sustained positive skill across the evaluated horizon range.

examined in this study,

$$\begin{aligned} H^*(\text{relax}) &= 48, & [h_{\text{start}}, h_{\text{end}}] &= [1, 48], \\ H^*(\text{strict}) &= 48, & H^*(\text{time}) &= 48 \text{ h.} \end{aligned} \tag{10}$$

Model-substitution check (wind).

Under the same protocol with ARIMA(2, 0, 0) and a true h -step forecast, the descriptor tuple is unchanged: $H^*(\text{relax}) = H^*(\text{strict}) = 48$, $[h_{\text{start}}, h_{\text{end}}] = [1, 48]$ and no sign changes. Skill amplitude differs from LightGBM.

9.3 Interpretation

Because the skill curve remains positive throughout the evaluated range, $H^*(\text{relax})$ and $H^*(\text{strict})$ coincide and equal H_{max} , and the maximizing interval is $[1, 48]$. Operationally, wind represents sustained positive skill: the model retains baseline-relative usefulness continuously across the entire evaluated horizon range.

10 Traffic Domain Results

10.1 Traffic experimental setting

The fourth domain considered is the METR-LA hourly traffic-speed dataset (Li et al. 2018), constructed from a univariate sensor trajectory. The experiment uses hourly observations, a persistence baseline, and the canonical comparable LightGBM protocol, with MAE-based skill evaluated for forecasting horizons up to $H_{\text{max}} = 72$. This domain

is especially suitable for studying fragmented skill profiles, in which predictive usefulness may arise in short, separated intervals rather than in a single sustained range.

10.2 Observed skill behavior

In the traffic domain, the skill curve is highly non-monotonic and fragmented under the canonical comparable LightGBM protocol (Figure 6): positive skill appears in short, separated intervals interspersed with negative regions. This shape is consistent with strong baseline competitiveness at many horizons, punctuated by intermittent recoveries where the model surpasses persistence. For LightGBM,

$$H^*(\text{relax}) = 72, \quad [h_{\text{start}}, h_{\text{end}}] = [46, 52], \quad H^*(\text{strict}) = 7, \quad H^*(\text{time}) = 7 \text{ h.} \quad (11)$$

The first positive interval occurs at $[17, 18]$, and the sign of $\text{Skill}(h)$ changes 19 times across $h \in \{1, \dots, 72\}$.

Model-substitution check (traffic).

Under the same protocol and an h -step forecast, ARIMA(2, 0, 0) gives

$$H^*(\text{relax}) = 72, \quad [h_{\text{start}}, h_{\text{end}}] = [38, 43], \quad H^*(\text{strict}) = 6, \quad H^*(\text{time}) = 6 \text{ h,} \quad (12)$$

with 13 sign changes across $h \in \{1, \dots, 72\}$. Relaxed reach is preserved, whereas contiguous usefulness and interval placement depend on the model.

10.3 Interpretation

These results illustrate why multiple descriptors are needed in fragmented settings. The relaxed descriptor records the last positive-skill horizon; its span covers all 72 horizons, of which 36 have non-positive skill. The strict descriptor and its location reveal that the longest continuous useful window is only seven hours and occurs at $[46, 52]$.

Table 3 presents the operational horizon descriptors derived for the four study domains under a common evaluation framework. Together, these four domains instantiate qualitatively different horizon-wise skill structures under a shared, leakage-free evaluation design, which is why a single horizon summary is insufficient. The profiles differ in reach, continuity, and interval placement; these are not mutually exclusive domain classes. Load’s primary range contains only one day. Traffic has short, separated positive intervals, while PM_{2.5} has early positive islands followed by a longer late interval. Wind maintains positive skill across its evaluated 48-hour range. Different horizon caps and temporal units prevent interpreting these summaries as a ranking of intrinsic domain predictability. Collectively, these findings indicate that fixed-horizon summaries can obscure substantial cross-domain variability in the structure of baseline-relative predictability.

10.4 Alternative-baseline sensitivity

Table 4 reports the fixed seasonal-persistence sensitivity analysis. Within this analysis, the LightGBM and baseline records share the same forecast origin, target timestamp,

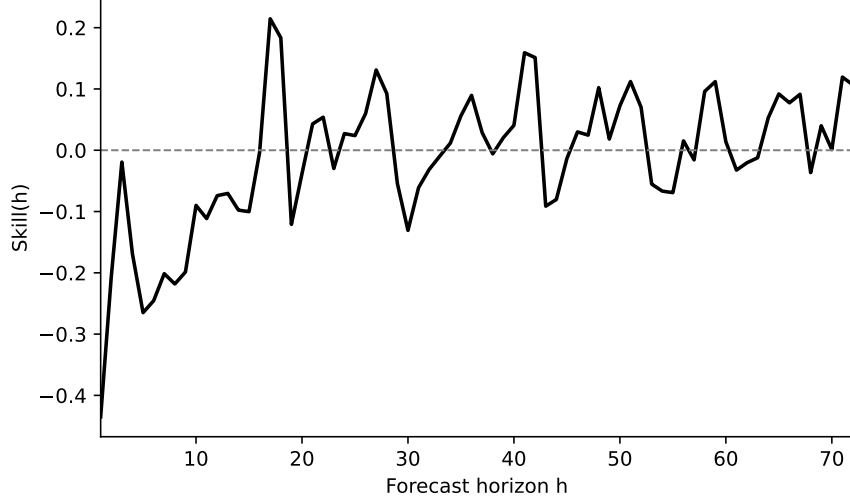


Fig. 6 Traffic skill curve $\text{Skill}(h)$ under the leakage-free rolling-origin protocol. The skill-profile pattern shows fragmented positive intervals: brief, separated stretches of positive skill interspersed with negative regions.

Domain	$H^*(\text{relax})$	$H^*(\text{strict})$	$[h_{\text{start}}, h_{\text{end}}]$	$H^*(\text{time})$	Notes
PM _{2.5}	48	22	[27, 48]	22 h	three positive intervals; the longest occurs late; five sign changes
Load (daily)	1	1	[1, 1]	1 day	positive skill is confined to the first evaluated day under this setup
Wind	48	48	[1, 48]	48 h	positive skill over the full evaluated range relative to persistence
Traffic	72	7	[46, 52]	7 h	fragmented profile with 19 sign changes and first positive interval [17, 18]

Table 3 Comparison of predictability horizons across domains (PM_{2.5}, electric load, wind, and traffic).

horizon, and y_{true} ; pairwise alignment loses no records. The PM_{2.5} matched-support persistence row and the primary LightGBM curve both yield 48/22 and [27, 48]. This agreement of horizon descriptors does not imply identical samples or MAE values. Baseline effects are assessed between the matched-support rows, not by comparing errors across different supports. The former primary value of 13 was a reporting mismatch, not an established effect of changing support.

Domain	Baseline	Common n_h	H_{relax}^*	H_{strict}^*	Longest interval
PM _{2.5}	Persistence	328–346	48	22	[27, 48]
PM _{2.5}	Seasonal persistence	328–346	48	24	[25, 48]
Load	Persistence	301	1	1	[1, 1]
Load	Seasonal persistence	301	1	1	[1, 1]
Wind	Persistence	350–354	48	48	[1, 48]
Wind	Seasonal persistence	350–354	48	30	[19, 48]
Traffic	Persistence	102–108	72	7	[46, 52]
Traffic	Seasonal persistence	102–108	72	14	[48, 61]

Table 4 Baseline sensitivity on exact common support. n_h is the number of aligned forecast origins at each horizon (a range is shown when it varies with h). The persistence rows in this table are recomputed on the same support as seasonal persistence and therefore need not equal the primary full-support persistence descriptors.

Relative to the matched persistence row, seasonal persistence changes H^* (strict) from 22 to 24 in PM_{2.5} and shifts the interval from [27, 48] to [25, 48]; load remains 1/1 at [1, 1]; wind changes from 48 at [1, 48] to 30 at [19, 48]; and traffic changes from 7 at [46, 52] to 14 at [48, 61]. In all four domains H^* (relax) is unchanged. For the hourly domains, however, the evaluated endpoint is an exact multiple of the 24-hour period, where seasonal persistence equals ordinary persistence; the unchanged relaxed endpoint is therefore partly structural and is not evidence of general baseline invariance. The changes in strict continuity and interval location demonstrate the substantive baseline conditionality of the descriptors.

11 Discussion

The four examples show information that a few fixed-horizon errors would miss. Traffic’s positive intervals are separated by non-positive regions. PM_{2.5} also has positive islands, but its longest interval occurs late and lasts 22 hours. Wind has positive skill throughout its evaluated range. Load has a one-day primary summary and a separate extended diagnostic. These descriptions concern the evaluated forecasts and baselines, not inherent limits of the domains.

Reach, continuity, and fragmentation can coexist within the same curve. An outer endpoint alone does not show the gaps before it, and an interval length alone does not locate that interval. Reporting both descriptors with interval locations makes these distinctions explicit without assigning each domain to a unique profile type. When skill curves are non-monotonic, total positive reach can remain stable even though contiguous usefulness and interval structure change across baselines and other components of the evaluation protocol.

Prior methodological work documents that leakage-prone evaluation and omitted strong baselines can distort conclusions about practical forecasting value (Hewamalage et al. 2023). Our results add that fixed-horizon reporting can also hide delayed or

fragmented usefulness, which the skill curve and its interval descriptors make visible. The persistence-relative rolling-origin design avoids leakage and baseline omission by construction, but it does not remove other sources of evaluation bias, such as the choice of metric or adaptive selection of horizons.

Potential dynamic use. An operational system could use an H^* profile estimated on historical validation data to freeze a set of eligible forecast horizons for subsequent deployment, and later update that policy using only errors that have already been realized. This paper does not validate such an adaptive controller. Re-estimating H^* , selecting horizons, and evaluating that choice on the same errors would introduce adaptive selection bias and overfitting. A dynamic policy therefore requires a prospective, prequential, or nested evaluation in which horizon selection is separated from the data used to assess it.

Metric choice. MAE and RMSE give different weight to large errors, so changing the loss changes the comparison summarized by H^* . Recomputing the descriptors from per-origin squared errors (Appendix A.1) leaves wind unchanged and $\text{PM}_{2.5}$ nearly so, but changes traffic and electric load materially. Cross-metric stability therefore cannot be assumed.

Baseline sensitivity. Seasonal persistence alters the longest continuous positive-skill window in $\text{PM}_{2.5}$, wind, and traffic, despite leaving relaxed reach unchanged in this design. Wind provides the clearest example: the strict interval contracts from the full $[1, 48]$ range under persistence to $[19, 48]$ under seasonal persistence. Traffic changes in the opposite direction, from a seven-hour to a fourteen-hour longest interval. These outcomes are reported without selecting the reference that gives a more favorable profile and show why the baseline is part of the estimand.

Scope of inference. The same forecasting model can have different horizon descriptors when the baseline, loss, evaluated range, or data changes. The descriptors summarize that comparison,

$$H^* = H^*(\text{dataset}, \text{model}, \text{baseline}, \text{metric}, \text{validation protocol}),$$

and do not measure an intrinsic limit of a model or domain. The alternative-baseline analysis tests sensitivity to the reference, not robustness across forecasting models.

These findings point to two methodological implications for horizon-based forecast evaluation:

1. Predictability should be assessed relative to an explicit baseline; in cross-domain studies, using a common baseline is often essential for comparability.
2. Horizon estimation should differentiate between relaxed and contiguous predictability, particularly when empirical skill curves are delayed or fragmented.

Overall, fixed-horizon summaries lose structural information that matters operationally: fragmented or delayed profiles can yield the same fixed-horizon summary while implying very different decision windows. Reporting $H^*(\text{relax})$ and $H^*(\text{strict})$ together (and interval structure when needed) separates total positive reach from sustained usefulness while making explicit that any H^* estimate is conditional on the baseline, metric, and evaluation protocol.

12 Limitations

The reported horizons apply to the datasets, forecasts, and evaluation choices studied here. Their values depend on the baseline, loss, maximum evaluated horizon, and temporal-validation design; they should not be interpreted as universal constants. The empirical scope is limited to four datasets and their stated temporal periods, the fixed LightGBM configurations reported above, and the evaluated horizon caps. Results need not transfer to another dataset, target, model configuration, baseline, metric, or $H_{\max}$. The alternative-baseline sensitivity examines baseline conditionality but does not eliminate it. Seasonal persistence also coincides with ordinary persistence at exact multiples of the seasonal period, so unchanged terminal reach at those endpoints is not an independent invariance result. Moreover, no dynamic H^* controller is evaluated prospectively; the operational policy described in the Discussion remains a potential application requiring separate prequential or nested validation. Positive sample-average skill is not, by itself, evidence of a material or statistically established operational benefit. We do not provide uncertainty intervals for the horizon descriptors or evaluate deployment costs.

13 Conclusion

This study frames predictability-horizon estimation as an operational forecast-evaluation task, focused on skill relative to an explicit baseline. The contribution is not a new forecasting model but a leakage-free, baseline-relative way to summarize useful predictive reach across horizons via the skill curve $\text{Skill}(h)$.

Using rolling-origin evaluation against persistence, we report two complementary horizon descriptors: $H^*(\text{relax})$ (the maximum evaluated horizon with any positive skill) and $H^*(\text{strict})$ (the longest uninterrupted run of positive skill), augmented when needed with interval locations and sign-change structure. Across the four domains, useful predictive reach differs not only in extent but also in continuity and fragmentation, and the baseline-sensitivity analysis makes explicit that these descriptors are conditional on the chosen reference. Future work could examine whether these features of skill profiles recur across broader domain families and evaluation designs, and whether more expressive model classes or multivariate formulations alter the interval structure of baseline-relative usefulness.

A.1 Metric sensitivity

The main analysis uses MAE-derived skill. To examine the effect of the loss, we recomputed $\text{Skill}(h) = 1 - \text{RMSE}_{\text{model}}(h)/\text{RMSE}_{\text{persistence}}(h)$ with the LightGBM models, persistence baseline, horizons and rolling-origin protocol of the main experiments unchanged. The MAE of this rerun reproduces the reported MAE values exactly in all four domains, so both metrics are computed on identical forecasts and support. Wind is unchanged and $\text{PM}_{2.5}$ keeps its late-recovery profile, with a slightly longer and earlier longest interval. In traffic the fragmented MAE profile (19 sign changes) becomes a long, mostly contiguous interval under RMSE (two sign changes), and in electric load positive skill extends from day 1 to day 3. Because RMSE weights

Domain	MAE			RMSE		
	H^* (relax)	H^* (strict)	Interval	H^* (relax)	H^* (strict)	Interval
PM _{2.5}	48	22	[27, 48]	48	23	[26, 48]
Load (1–7 d)	1	1	[1, 1]	3	3	[1, 3]
Wind	48	48	[1, 48]	48	48	[1, 48]
Traffic	72	7	[46, 52]	72	63	[10, 72]

Table 5 Horizon descriptors under MAE and RMSE for the same LightGBM forecasts. Load is evaluated over the seven-day auxiliary range; its primary summary uses one day.

large errors more heavily, the choice of loss is part of the operational definition of the descriptors.

A.2 ARIMA order diagnostic

The order $(2, 0, 0)$ used in the substitution checks was fixed a priori. To document how a standard procedure would behave on the same data, we ran `auto_arima` (AIC, $d = 0$, $p, q \leq 5$, with constant) on the $h = 1$ training window of 20 evenly spaced evaluated origins per domain (windows of 216–720 observations), using the windows of the rolling-origin protocol. No refitting of the reported check was performed with these orders. The selected order is not stable across origins in either domain, and $(2, 0, 0)$

Domain	Distinct best orders	Median Δ AIC of $(2, 0, 0)$	Max Δ AIC	Δ AIC ≤ 2
Wind	10	3.7	34.3	10/20
Traffic	8	28.0	68.5	4/20

Table 6 AIC-selected order across 20 origins per domain. Δ AIC is the AIC of $(2, 0, 0)$ minus the best AIC among the candidates. The most frequent best orders are $(1, 0, 0)$ for wind (6/20) and $(3, 0, 1)$ for traffic (5/20).

is generally further from the AIC optimum in traffic than in wind. Figures 7 and 8 show the ACF and PACF of the last evaluated training window of wind and traffic, respectively (720 observations, lags up to 48). In both series the PACF is dominated by lag 1 (0.80 in wind, 0.85 in traffic) with small coefficients at lags 2–4 (Figures 7 and 8, right panels). The traffic ACF also has a daily peak at lag 24 (0.36; Figure 8, left panel) that a non-seasonal AR(2) cannot represent. These plots were inspected after the order was fixed, not used to select it. The ADF test does not reject a unit root in 2 of 20 wind windows and 1 of 20 traffic windows ($p > 0.05$).

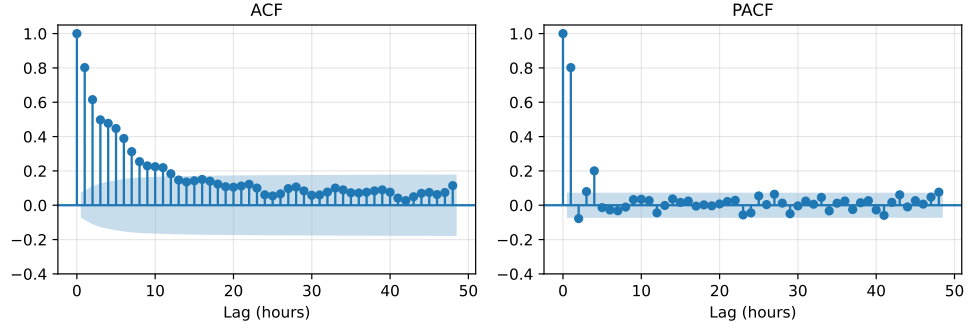


Fig. 7 Wind: ACF (left) and PACF (right) of the last evaluated $h = 1$ training window (720 hourly observations). The shaded band is the 95% confidence region.

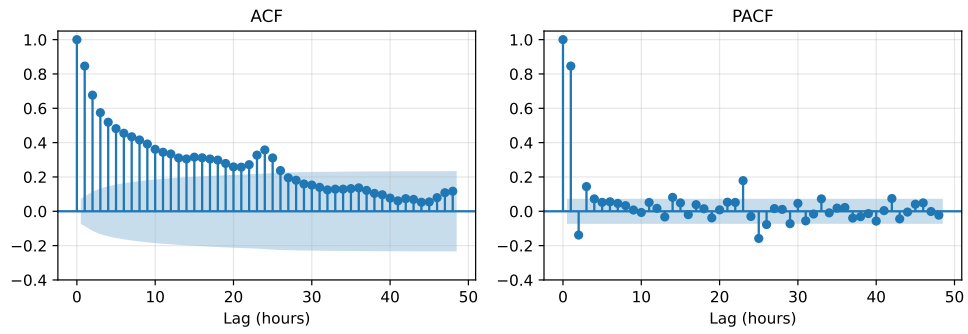


Fig. 8 Traffic: ACF (left) and PACF (right) of the last evaluated $h = 1$ training window (720 hourly observations). The shaded band is the 95% confidence region.

Declarations

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Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics Approval

Not applicable. This study is based exclusively on publicly available benchmark datasets. It did not involve human participants, animals, patient data, or primary data collection.

Data Availability

All datasets used in this study are publicly available. The electric load dataset is available from the UCI Machine Learning Repository (Trindade 2015), accessible at <https://doi.org/10.24432/C58C86>; we use the daily sum of all 370 client series. The wind speed data are from NREL’s Wind Toolkit (WTK-LED CONUS, 2019; one grid point, 100 m wind speed) (Draxl et al. 2015). The PM_{2.5} dataset is the Beijing PM_{2.5} dataset archived by the UCI Machine Learning Repository (Chen 2015), available at <https://doi.org/10.24432/C5JS49>. The traffic benchmark corresponds to the METR-LA dataset, originally introduced by Li et al. (2018); the processed traffic file (`metr-la.h5`) was obtained from the public data links provided in the original DCRNN repository, and we use the hourly mean of sensor 773869. The exact input files used for all four experiments, with SHA-256 checksums, are deposited at https://github.com/EduIntellect/paper2H/tree/data-cache-v1/release/data_cache_four_domain. The traffic measurements originate from Caltrans PeMS; we have not obtained separate confirmation of their redistribution terms and will remove that file on request.

Code Availability

The replication package accompanying this article contains the data-processing, forecasting, common-support validation, descriptor, table, and figure-generation scripts used for the reported results. The public project location is <https://github.com/EduIntellect/paper2H>.

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